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AUGUSTO FELIPE MAGGIONI

OBJECTIVE

Seeking a role as a Data Scientist / AI Engineer / Machine Learning Engineer, designing and deploying solutions based on LLMs, RAG and machine learning in production environments, integrated with data engineering and backend systems. I am looking for dynamic, innovation-driven environments with highly complex challenges, where I can continuously expand my knowledge, apply AI in practice and deliver real, measurable business impact.

PROFESSIONAL EXPERIENCE

10/2024 - Present

Data Scientist

2 years and 1 month

Vert Analytics - Brasília, Federal District

- Developed data and AI solutions to automate legal processes, increasing operational efficiency and business value
- Built data pipelines and developed machine learning models to extract and analyze relevant information
- Implemented LLM (Gemini and open-source models) and RAG solutions with LangChain and LangGraph, with semantic search on pgvector, Qdrant, Chroma, FAISS and Elasticsearch, to read, interpret and generate documents
- Applied RAG with similarity search and information retrieval to build a chatbot
- Implemented OCR with Tesseract, Chandra and Docling to extract text from documents and structure the data
- Used AWS Bedrock, Azure Document Intelligence and OpenRouter
- Set up evaluation for the LLM/RAG systems: a reference set for regression testing, retrieval metrics (MRR, precision and recall), text generation metrics (ROUGE and BLEU), expert human review and cost and latency monitoring
- Used MLflow to track experiments and version models
- Built REST APIs with FastAPI to deliver the AI solutions
- Automated critical processes, cutting manual document review time by **80%+**
- Supported decision-making in the legal and operational areas through data-driven solutions

Technologies: Python, Jupyter Notebook, Machine Learning, LLMs, Gemini, RAG, LangChain, LangGraph, AWS Bedrock, OpenRouter, MLflow, NLP, OCR, Tesseract, Chandra, Docling, Azure Document Intelligence, pgvector, Qdrant, Chroma, FAISS, Elasticsearch, FastAPI, REST APIs, Backend, PostgreSQL/SQL, Docker, Kubernetes

12/2023 - 11/2024

Full-Stack Data and Artificial Intelligence Developer

1 year

M9 Inteligência em Mobilidade - Passo Fundo, RS

- Developed an event-driven (SSE) Python backend for a tow truck management system, processing real-time vehicle and customer data
- Built data pipelines and integrated machine learning models into production, reducing data processing and analysis time
- Applied OCR to extract text from documents and structure the data
- Implemented RAG with LangChain and LangGraph, using similarity search on Qdrant to retrieve similar information
- Built APIs with FastAPI

- Optimized PostgreSQL queries and data modeling, improving application performance and scalability
- Contributed to a distributed systems architecture with Redis, Kafka and Docker, ensuring high availability and resilience

Technologies: Python, Jupyter Notebook, FastAPI, PostgreSQL, Redis, Kafka, Docker, Linux, C#, Meilisearch, Databricks, TensorFlow, Scikit-learn, BERT, LLMs, RAG, LangChain, LangGraph, Qdrant, OCR

05/2023 - 11/2024

Back-end Developer

1 year and 7 months

M9 Inteligência em Mobilidade - Passo Fundo, RS

- Developed and maintained Python backend services for distributed processing of millions of daily records from vehicle trackers
- Implemented integrations with tracking devices and onboard video systems, ensuring continuous data ingestion and processing
- Built APIs with FastAPI
- Optimized algorithms and data pipelines, improving performance and efficiency in large-scale processing
- Designed and managed a SQL Server database, ensuring consistency, performance and availability
- Managed servers and infrastructure (Linux/AWS EC2), ensuring high availability and reliability of applications
- Contributed to a distributed systems architecture using Redis, Kafka and Docker

Technologies: Python, Jupyter Notebook, FastAPI, SQL Server, Redis, Kafka, Docker, Linux, AWS EC2, MongoDB, Postman, Networking

06/2021 - 12/2023

Data Scientist and Data Engineer

2 years and 7 months

Cooperando - Passo Fundo, RS

- Developed end-to-end data pipelines (extraction, processing and storage) using web scraping, DataFrames and databases such as MongoDB, DynamoDB and MySQL
- Structured and processed large volumes of data with Pandas, NumPy and Scikit-learn, enabling analyses and business insights
- Performed data analysis with Power BI, Excel and Python, supporting data-driven decision-making
- Applied OCR to extract text from documents and structure the data
- Implemented RAG with LangChain, using text similarity with BERT embeddings to retrieve similar information
- Developed and evaluated machine learning models for classification and prediction (Naive Bayes, Decision Trees, K-Means, Regression, SVM and Neural Networks)
- Implemented a sentiment analysis model for Portuguese with BERT Large, classifying into 10 classes and trained on a proprietary dataset
- Built APIs with FastAPI
- Deployed applications on AWS (EC2, S3, DynamoDB, Lambda) with Flask and Nginx, ensuring scalability and on-demand cost

Technologies: Python, Jupyter Notebook, FastAPI, RAG, LangChain, OCR, Pandas, NumPy, Scikit-learn, BERT, Flask, Nginx, AWS (EC2, S3, DynamoDB, Lambda), MongoDB, MySQL, Power BI, Excel

EDUCATION

02/2024 - 03/2026

Master's Degree in Applied Computing: Data Science and Artificial Intelligence
University of Passo Fundo - Passo Fundo, RS
Status - Completed

02/2018 - 12/2023

Bachelor's Degree in Computer Science
University of Passo Fundo - Passo Fundo, RS
Status - Completed

SKILLS

- Professional ethics and confidentiality
- Development of **Artificial Intelligence (AI) and Machine Learning** solutions in production
- Building **data engineering pipelines (ETL/ELT)** for large-scale processing
- Implementation of applications with **LLMs and RAG (Retrieval-Augmented Generation)**, LangChain and LangGraph
- Semantic search with **vector databases** (pgvector, Qdrant, Chroma, FAISS and Elasticsearch)
- Document extraction with **OCR** (Tesseract, Chandra, Docling and Azure Document Intelligence)
- Development of **REST APIs** with FastAPI and **scalable backend services**
- Database modeling and optimization (**PostgreSQL, SQL Server, MongoDB, DynamoDB and Neo4J**)
- Distributed systems architecture (**Kafka, Redis, Docker**)
- Deployment and management of applications in the **cloud (AWS EC2, S3, Lambda, Bedrock)**
- Experimentation with **MLflow** and **Jupyter Notebook**

SCIENTIFIC PUBLICATIONS

Cardiac Arrhythmia Detection: A Lead I Approach with Hybrid Neural Networks
Revista Interfaces: Saúde, Humanas e Tecnologia

Development of a hybrid neural network model for detecting cardiac arrhythmias from ECG signals (lead I). The work involves biomedical signal processing, feature extraction and the application of machine learning techniques to classify cardiac patterns, focusing on accuracy and applicability in real-world scenarios.
Link: https://interfaces.unileao.edu.br/index.php/revista-interfaces/article/view/1875

PERSONAL WEBSITES

Radar Eleitoral
Brazilian elections data dashboard - independent personal project

Web dashboard with Brazilian election results from 2010 to 2026, built from the open data of the Superior Electoral Court (TSE) and broken down by state, municipality and polling station. It brings together interactive maps and charts, comparisons across elections, live vote-count tracking and an LLM-based assistant that answers questions about the data through a tool-using agent and SQL queries over its own databases.
Technologies: Python, Dash, Plotly, DuckDB, PostgreSQL, Pandas, PyArrow, LangChain, LLMs, Docker
Link: https://radar-eleitoral.oladata.com.br/

Portfolio
Data science, AI and data engineering projects (GitHub Pages)
Link: https://augustobook.github.io/portfolio/